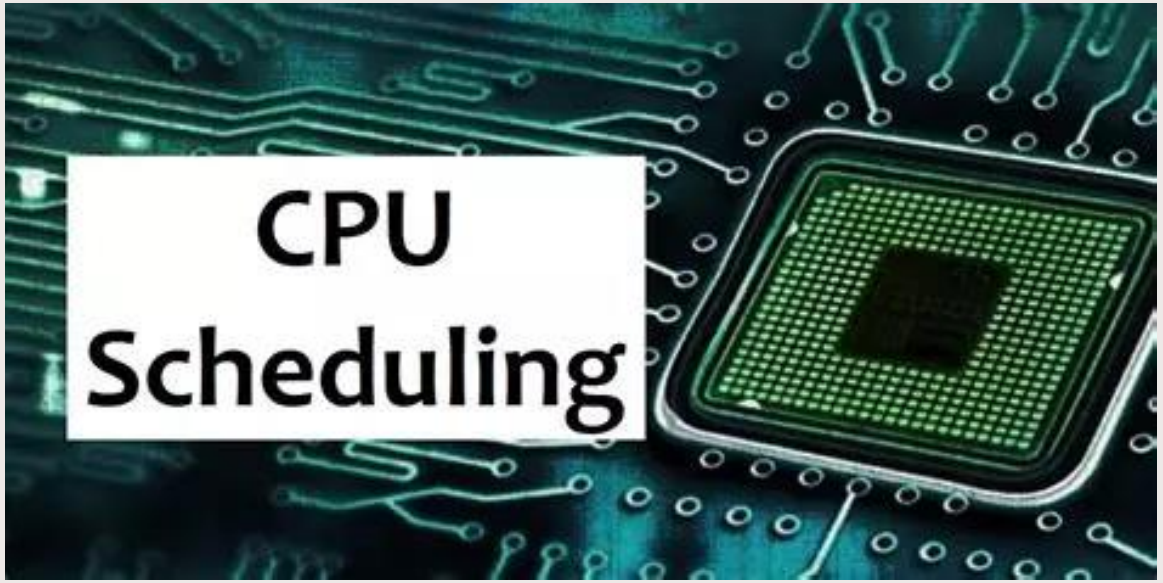


FCFS SCHEDULING & BANKER'S ALGORITHM

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INTRODUCTION

- OS Algorithm Simulator was used on Android
- Simulated CPU scheduling algorithms
- Simulated deadlock avoidance algorithms
- Waiting time and turnaround time were calculated
- Banker's Algorithm was used to test safe states

PROJECT OBJECTIVES

1

Part A:

- Implement FCFS scheduling
- Analyse waiting time and turnaround time
- Observe CPU execution order

2

Part B:

- Implement Banker's Algorithm
- Calculate Need Matrix
- Determine safe sequence

FIRST COME FIRST SERVED (FCFS)

Non-preemptive scheduling algorithm

Processes execute according to arrival time

First arriving process gets CPU first

Process runs until completion

Simple and easy scheduling method

PROCESS CONFIGURATION

PID	Arrival	Burst	Priority
P1	0	8	2
P2	2	6	1
P3	4	10	3
P4	6	5	4
P5	8	7	2
P6	10	9	5
P7	12	5	1
P8	14	6	6
P9	18	8	3
P10	18	5	7
P11	20	7	4
P12	22	5	2

FCFS SIMULATION STEPS



1. P1 EXECUTES FIRST AT TIME 0



2. PROCESSES ENTER READY QUEUE



3. CPU EXECUTES ACCORDING TO ARRIVAL ORDER



4. EACH PROCESS WAITS FOR PREVIOUS PROCESS



5. WAITING TIME INCREASES FOR LATER PROCESSES

SCHEDULING FORMULAS



Turnaround Time (TAT)



= Finish Time − Arrival Time



Waiting Time (WT)



= Turnaround Time − Burst Time

FCFS RESULTS

PID	Finish	TAT	WT
P1	8	8	0
P2	14	12	6
P3	24	20	10
P4	29	23	18
P5	36	28	21
P6	45	35	26
P7	50	38	33
P8	56	42	36
P9	64	48	40
P10	69	51	46
P11	76	56	49
P12	81	59	54

DISCUSSION — FCFS

- Processes executed in arrival order
- Waiting time increased for later processes
- Long processes delayed smaller processes
- Convoy effect was observed
- FCFS is simple but not efficient

BANKER'S ALGORITHM

Deadlock avoidance algorithm

Checks if system is safe

Uses:

- Maximum Matrix

- Allocation Matrix

- Available Vector

- Need Matrix

NEED MATRIX CALCULATION

- Need = Maximum – Allocation

Process	R1	R2	R3
P1	7	4	3
P2	1	2	2
P3	6	0	0
P4	0	1	1

SAFE SEQUENCE

Work = [3,3,4]

Safe Sequence:

P2 → P4 → P1 → P3

The system is in a safe state.

CONCLUSION

- FCFS scheduling was implemented successfully
- Waiting time increased for later processes
- Banker's Algorithm confirmed safe execution
- Simulation improved understanding of OS concepts